HOTEL REVENUE ANALYTICS

Hotel Booking Pace and Pickup Analysis

How far ahead guests book, how demand builds toward arrival, and what drives the busiest dates at a City Hotel and a Resort Hotel

75,166	45 days	23.7%	14.1%
stayed bookings analysed	median booking window	booked within 7 days of arrival	booked more than 180 days ahead

About this report

This report analyses booking pace and pickup for two hotels over arrival dates from July 2015 to August 2017. Using the date each booking was made (derived from its lead time), it shows how demand builds before arrival, how that build differs by hotel, weekday, and market segment, how rates vary with lead time, and why the ten busiest arrival dates reached their peaks.

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SECTION 01

Executive summary

Bookings are made late, on very different clocks by hotel, weekday and segment, and the busiest dates are decided by single large block bookings.

1

Half of bookings come within 45 days

The median booking is made 45 days before arrival (mean 80). 23.7% are made within a week and 7.9% on the day of arrival.

2

The long tail still matters

14.1% of bookings are made more than 180 days ahead: 36.6% of them from Offline TA/TO, 32.8% from Online TA and 24.4% from Groups.

3

Resort books later than City

Resort median lead time is 38 days against 50 for City, and 10.6% of Resort bookings are made on the arrival day against 6.1% at City.

4

Weekend arrivals book earlier

Fri to Sun arrivals have a median lead time of 51 days against 41 for Mon to Thu, and only 5.8% are same-day against 9.4%.

5

Segments run on different clocks

Median lead time ranges from 6 days for Corporate and 11 for Direct to 90 for Offline TA/TO and 107 for Groups.

6

Late bookings carry lower rates

Same-day bookings average an ADR of EUR 83, against EUR 104 to 106 for bookings made 8 to 180 days ahead.

7

Pickup concentrates in the final days

For August 2017 arrivals, 15.0% of bookings were made within 7 days. Four of the five densest heatmap cells fall within 2 days of arrival.

8

Block bookings set the peaks

On six of the ten busiest dates, one block made on a single day accounted for 38% to 57% of the date's bookings.

Data and method

- **Dataset:** Hotel Booking Demand (Antonio, Almeida and Nunes, 2019), with 119,390 bookings and 32 columns for a City Hotel and a Resort Hotel in Portugal.
- **Arrival date** was built from the arrival year, month and day. **Booking date** was derived as arrival date minus lead time, because the dataset contains no booking timestamp.
- **Stayed bookings only:** bookings with reservation status Check-Out were kept (75,166 of 119,390). The 43,017 cancellations and 1,207 no-shows are excluded, so all results describe bookings that were honoured.
- **Pace curves and pickup tables** were built by grouping on arrival date and lead time in days. Weekend arrivals are Friday to Sunday; weekday arrivals are Monday to Thursday.
- **ADR** is the average daily rate recorded per booking. Booking counts refer to booking records, not rooms.

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SECTION 02

Booking window

The typical guest books about six weeks ahead, but demand is spread across a very long window.



Figure 1. Distribution of lead time for 75,166 stayed bookings (50 bins).

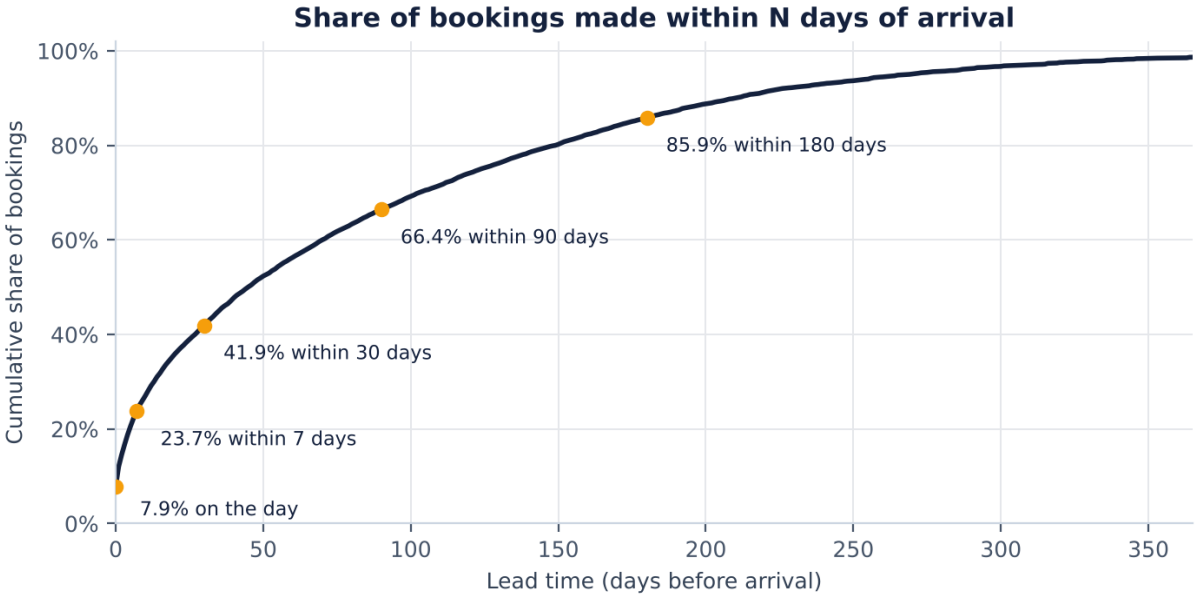


Figure 2. Cumulative share of bookings made within N days of arrival.

45 days	80 days	5,915	737 days
median lead time	mean lead time	same-day bookings (7.9%)	longest booking window

- Lead times are strongly right-skewed: the mean (80 days) sits far above the median (45 days) because a minority of bookings are made many months ahead.
- 23.7% of bookings are made within 7 days, 41.9% within 30 days, 56.3% within 60 days and 66.4% within 90 days. 85.9% are made within 180 days and 98.6% within a year.

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SECTION 03

Pace curve of the busiest arrival date

The busiest date, 5 December 2015, was built from three large Groups blocks rather than from steady daily pickup.

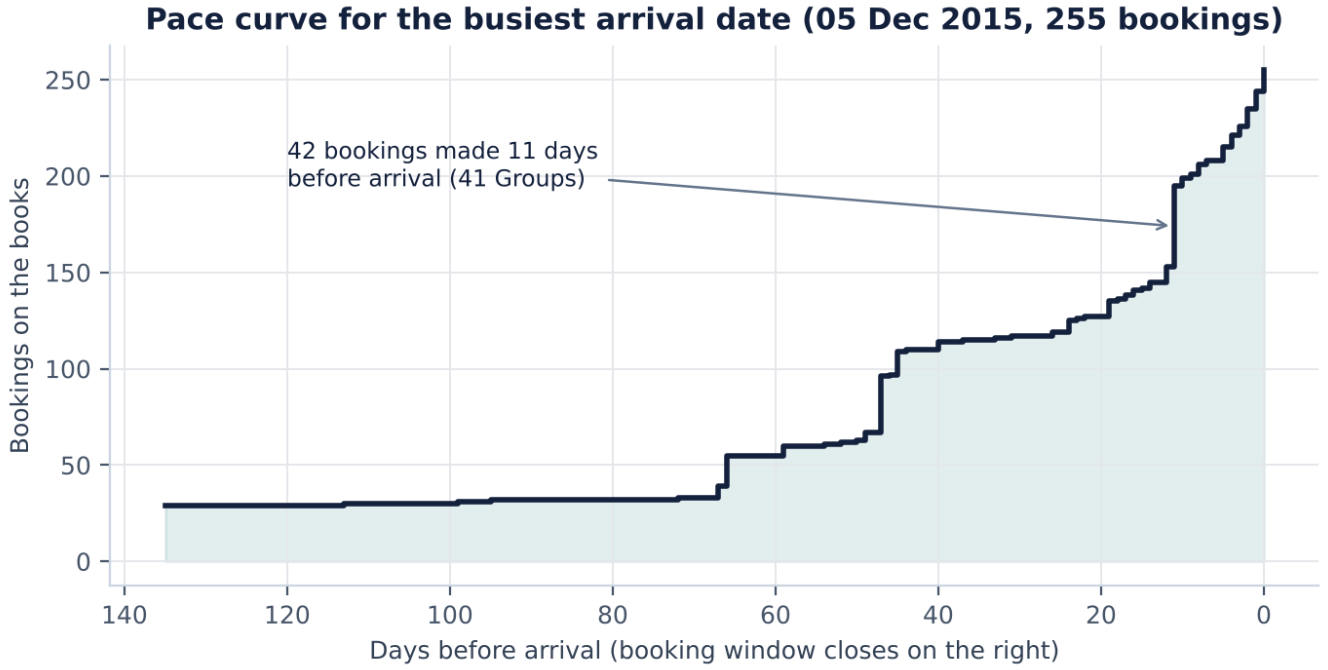


Figure 3. Cumulative bookings on the books by days before arrival, 5 December 2015 (255 stayed bookings).

255	19 days	42	21.6%
bookings on the busiest date	median lead time (45 for all bookings)	bookings made 11 days out	on the books 60 days out (44.1% overall)

- The curve moves in three steps: 29 bookings made 135 days out (all Groups, City Hotel), 29 bookings made 47 days out (28 Groups) and 42 bookings made 11 days out (41 Groups, split across both hotels).
- Compared with all bookings, this date was built late. At least 60 days before arrival only 21.6% of its final bookings were on the books, against 44.1% for the full dataset. At 30 days the figures are 45.9% and 58.6%.
- Between those marked steps the curve rises slowly, and the final week adds about 47 bookings (from 208 on the books at 7 days to 255 at arrival).

Takeaway. A single pace curve for a peak date is dominated by when its group blocks were contracted, so forecasts for such dates should track block pickup separately from transient pickup.

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SECTION 04

Pickup heatmap, August 2017

Almost every arrival date in August 2017 shows its heaviest pickup in the last few days before arrival.

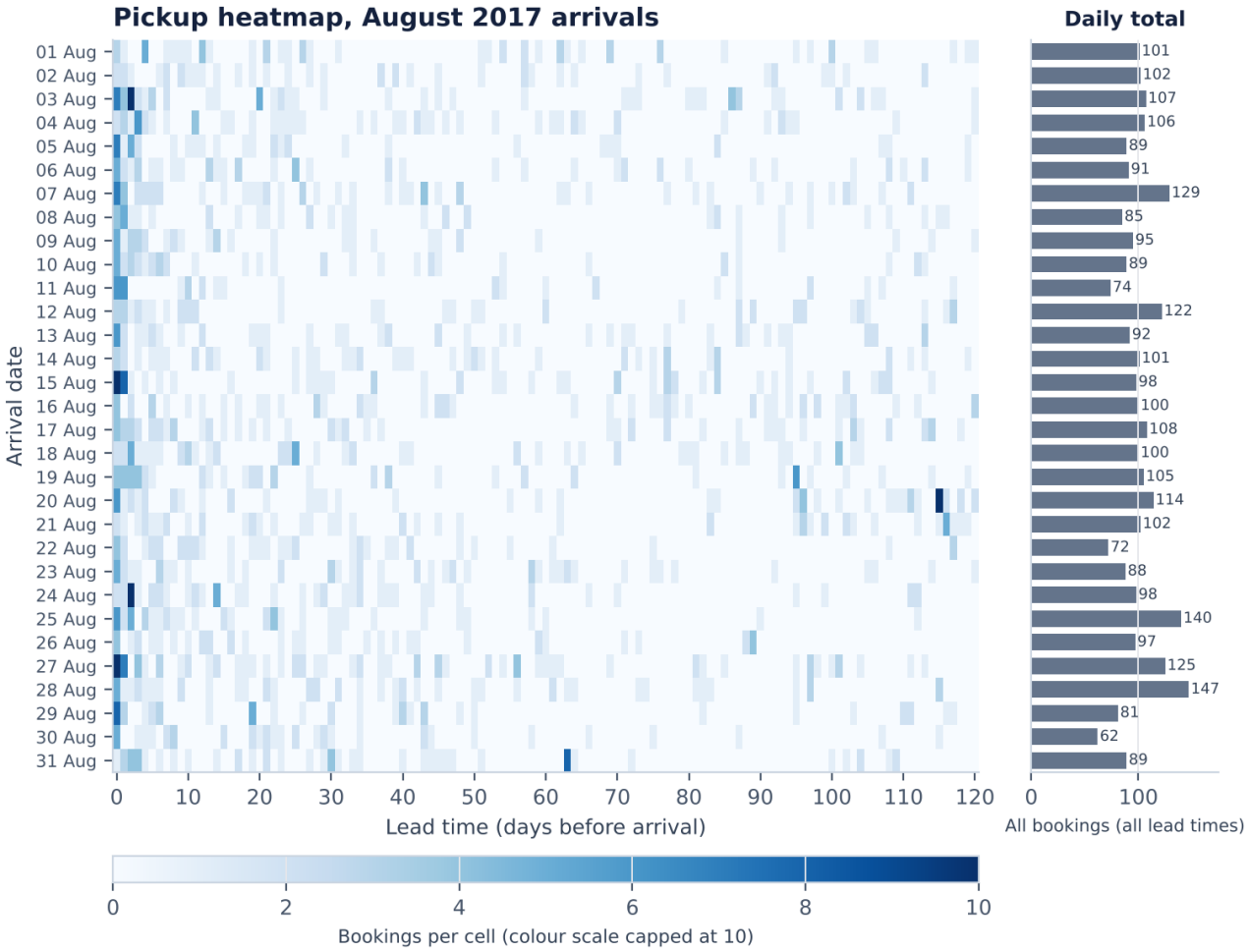


Figure 4. Bookings by arrival date and lead time (up to 120 days). Colour scale capped at 10 bookings per cell; bars show each date's total bookings across all lead times.

- August 2017 contains 3,109 stayed bookings (City 2,002, Resort 1,107), an average of 100 per arrival date with a range of 62 to 147.
- 11.2% of these bookings were made within 3 days of arrival, 15.0% within 7 days and 51.4% within 120 days.
- Five cells reach 10 or more bookings: 24 Aug at 2 days out (14), 3 Aug at 2 days (12), 27 Aug on the day (12), 15 Aug on the day (11) and 20 Aug at 115 days out (10). The two cells at 2 days out are mostly Groups bookings (11 of 14 and 8 of 12), and the early cell on 20 Aug is a single Offline TA/TO block of 10 bookings at the City Hotel.

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SECTION 05

City Hotel vs Resort Hotel

Both hotels peak on the arrival day, but Resort bookings are concentrated closer to arrival.

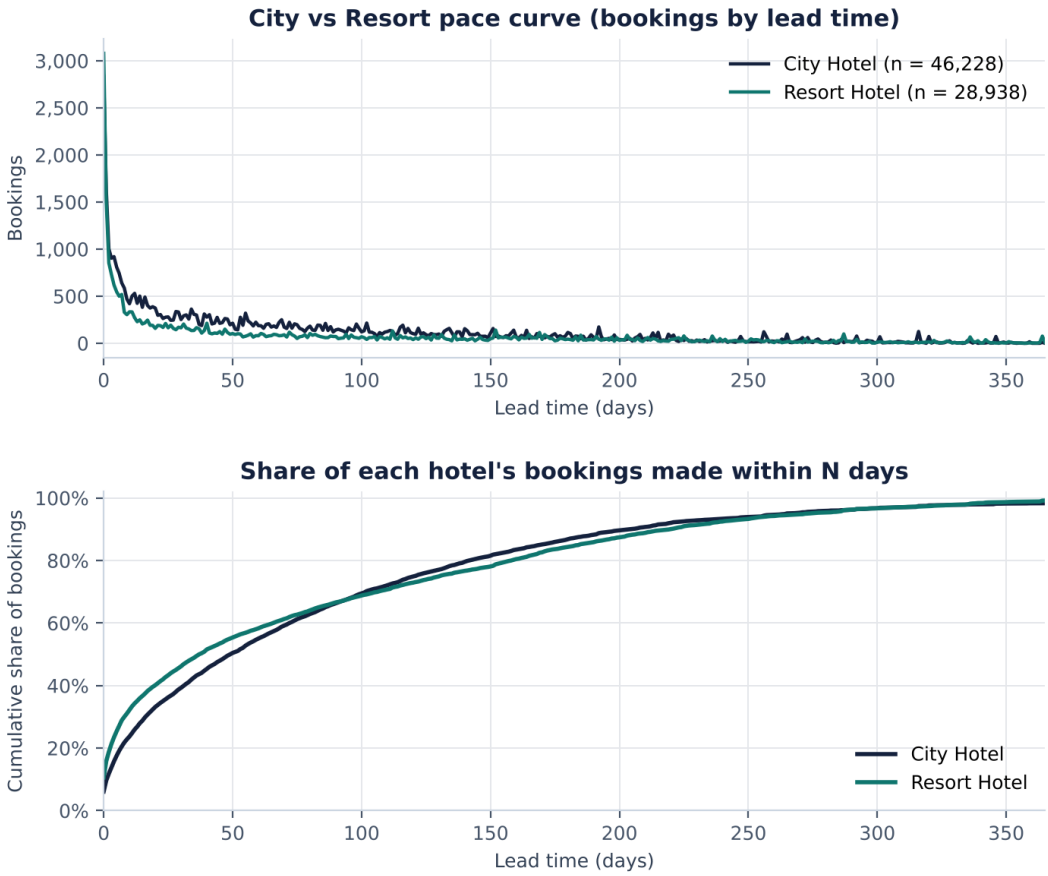


Figure 5. Top: bookings by lead time (raw counts, up to 365 days). Bottom: cumulative share of each hotel's own bookings.

Metric	City Hotel	Resort Hotel
Stayed bookings	46,228	28,938
Median lead time	50 days	38 days
Mean lead time	80.7 days	78.8 days
Same-day bookings	6.1%	10.6%
Within 7 days	20.5%	28.9%
Within 30 days	39.3%	46.2%
Within 90 days	66.3%	66.6%
More than 180 days	13.2%	15.6%
Average ADR	EUR 105.7	EUR 90.8

Takeaway. In raw counts City sits above Resort at most lead times because it holds 61.5% of stayed bookings. On a share basis Resort is later: 28.9% of its bookings arrive within a week against 20.5% at City, and City earns a higher average rate.

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SECTION 06

Weekend vs weekday arrivals

Leisure-weighted weekend arrivals book about ten days earlier than weekday arrivals.

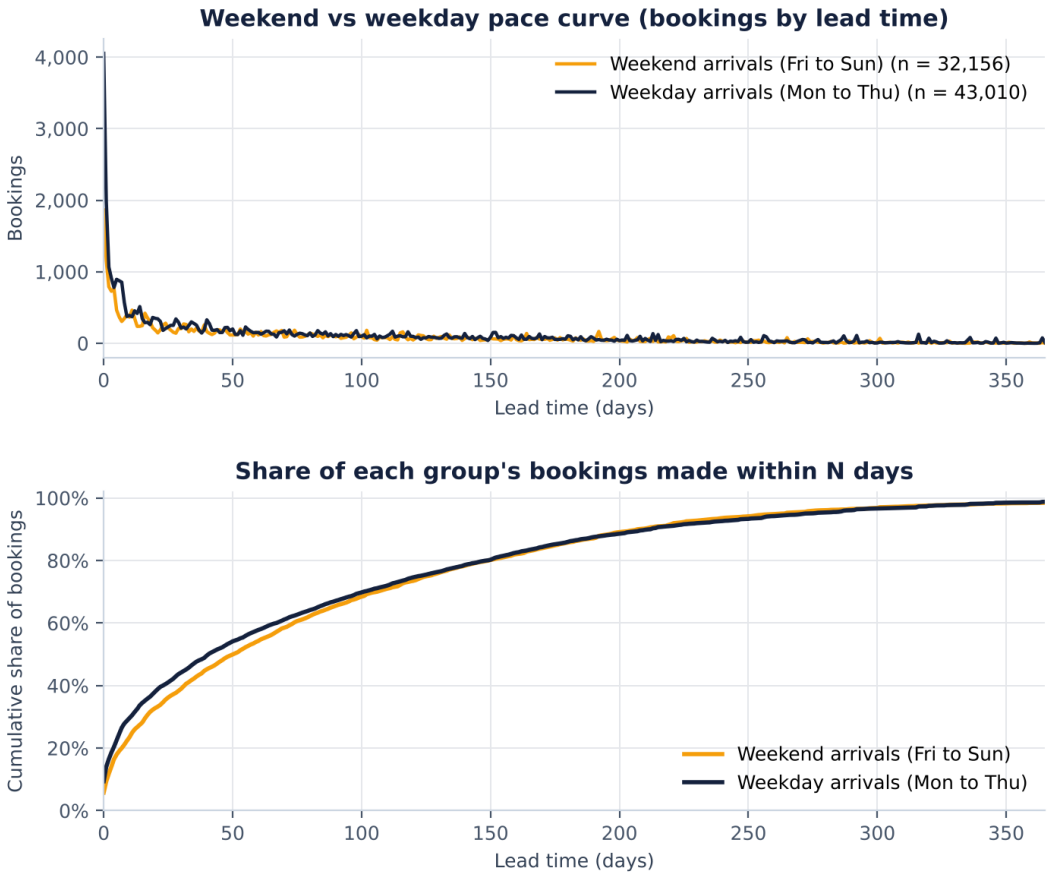


Figure 6. Top: bookings by lead time (raw counts). Bottom: cumulative share of each group's own bookings.

Metric	Weekend (Fri to Sun)	Weekday (Mon to Thu)
Stayed bookings	32,156	43,010
Median lead time	51 days	41 days
Mean lead time	82.2 days	78.3 days
Same-day bookings	5.8%	9.4%
Within 7 days	20.0%	26.5%
Within 30 days	39.0%	44.2%
More than 180 days	14.3%	14.0%

Takeaway. Weekday arrivals have more bookings in raw counts because they cover four days of the week against three. Normalised by group size, weekday demand is the more last-minute: 26.5% arrives within a week against 20.0% for weekends.

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SECTION 07

Market segment pickup

Each market segment books on its own clock, from Corporate and Direct bookings close to arrival to Groups and travel agents far ahead.

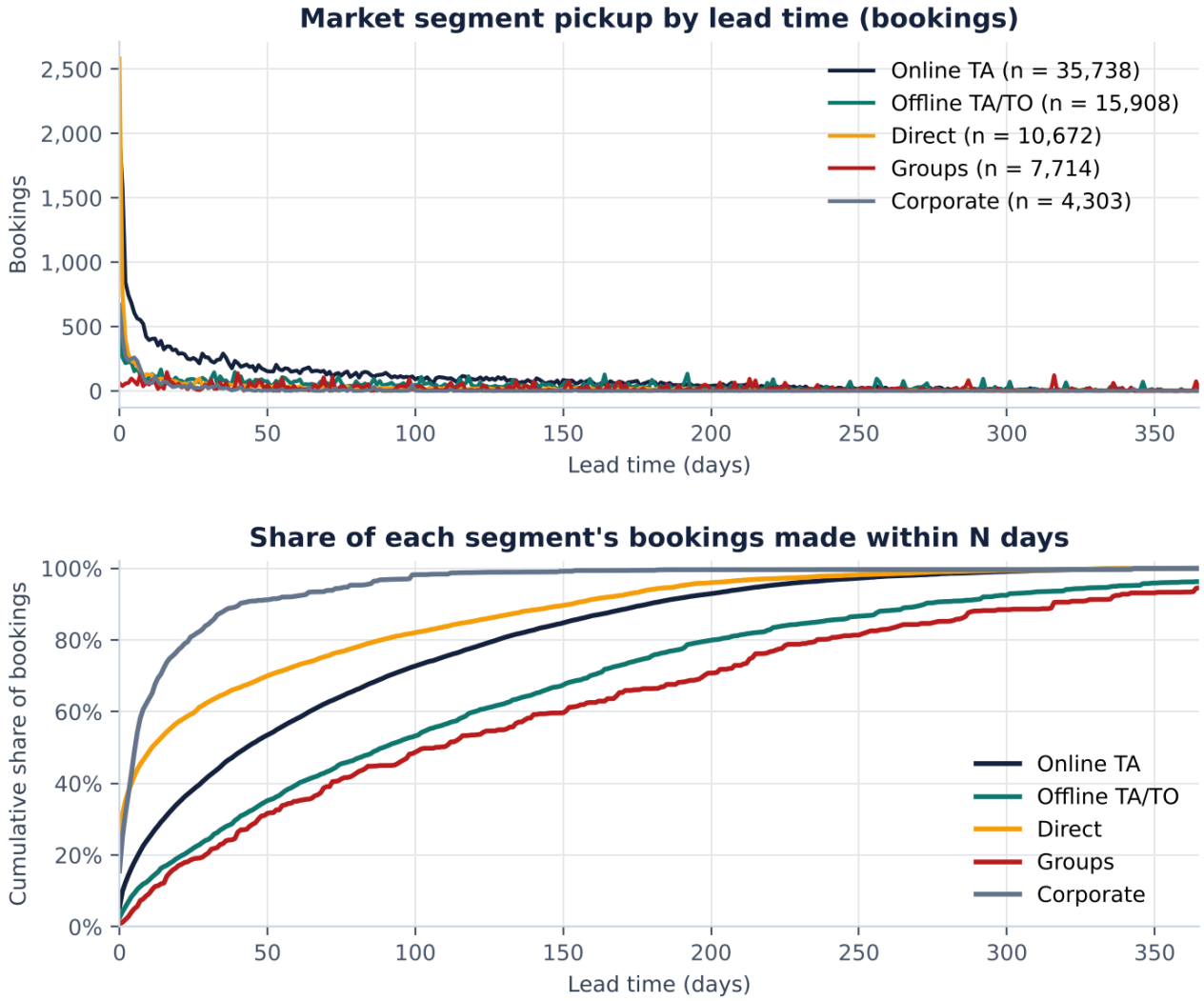


Figure 7. Top: bookings by lead time for the five largest segments (98.9% of bookings). Bottom: cumulative share of each segment's own bookings.

- **Online TA** is the largest segment (47.5%) and has both a heavy last-minute component and a long, thick tail of early bookings.
- **Direct** and **Corporate** book late: 45.3% and 58.2% of their bookings arrive within a week, and 24.2% of Direct bookings are made on the arrival day.
- **Offline TA/TO** and **Groups** book early: 24.4% and 33.6% of their bookings are made more than 180 days ahead, with medians of 90 and 107 days.

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SECTION 08

Market segment profile

Median lead time separates the segments more clearly than any other measure.

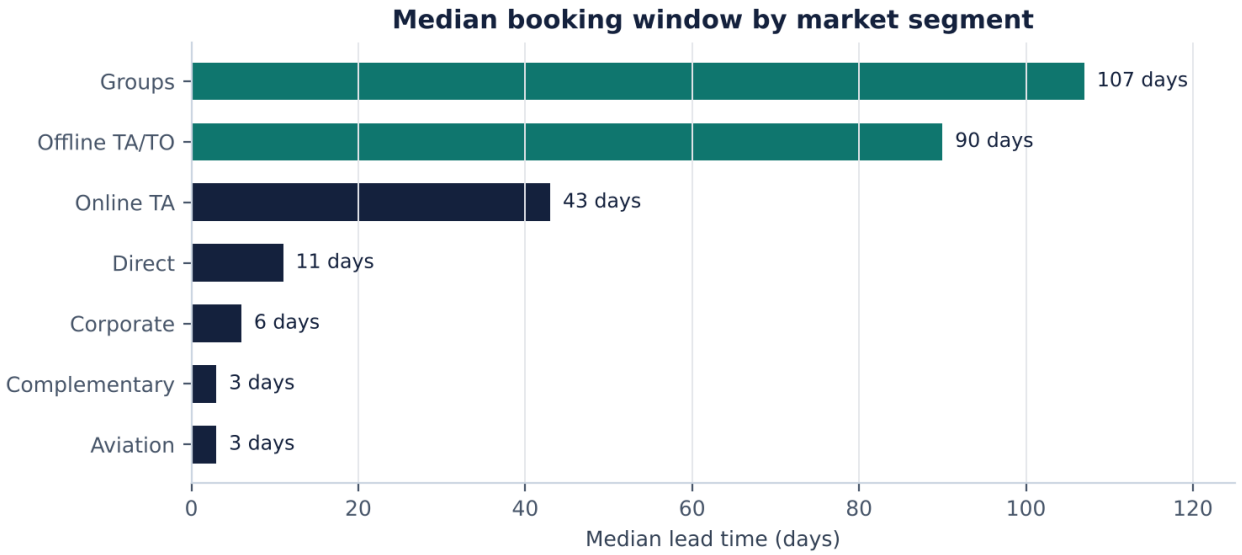


Figure 8. Median lead time by market segment.

Segment	Bookings	Share	Median lead	Same-day	Within 7 days	Over 180 days	Avg ADR
Online TA	35,738	47.5%	43 days	5.5%	21.1%	9.7%	EUR 113.9
Offline TA/TO	15,908	21.2%	90 days	2.6%	11.2%	24.4%	EUR 83.4
Direct	10,672	14.2%	11 days	24.2%	45.3%	5.9%	EUR 113.9
Groups	7,714	10.3%	107 days	0.8%	7.3%	33.6%	EUR 77.1
Corporate	4,303	5.7%	6 days	15.6%	58.2%	0.6%	EUR 67.1
Complementary	646	0.9%	3 days	28.8%	70.0%	0.6%	EUR 3.1
Aviation	185	0.2%	3 days	17.8%	82.2%	0.0%	EUR 99.5

- Of the 17,844 bookings made within 7 days, 42.3% come from Online TA, 27.1% from Direct, 14.0% from Corporate and 10.0% from Offline TA/TO.
- Of the 10,615 bookings made more than 180 days ahead, 36.6% come from Offline TA/TO, 32.8% from Online TA and 24.4% from Groups.
- Online TA and Direct share the highest average rate (EUR 113.9); Groups and Corporate book at lower average rates (EUR 77.1 and EUR 67.1).

Takeaway. Because medians range from 3 to 107 days, a single pace curve for the whole hotel blends very different booking behaviours. Segment-level curves describe the demand build far more accurately.

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SECTION 09

ADR by lead time

Bookings made on the arrival day carry the lowest average rate; rates are highest for bookings made one to six months ahead.

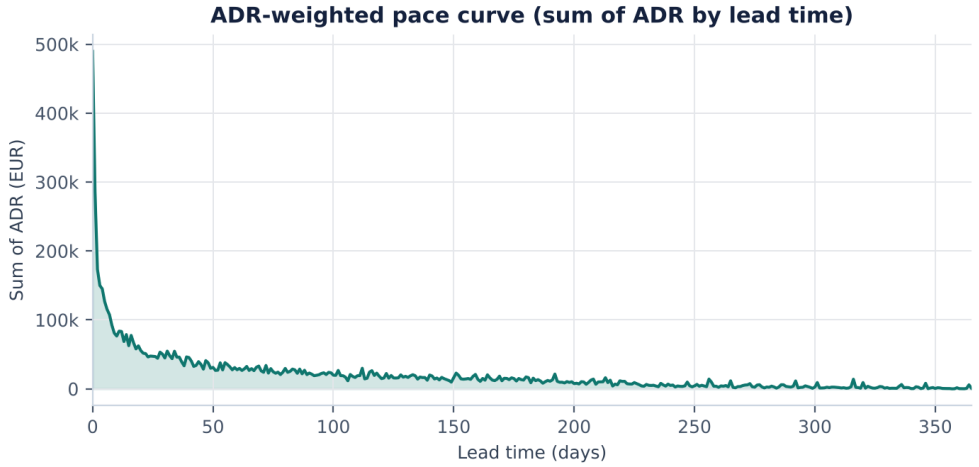


Figure 9. ADR-weighted pace curve: the sum of ADR across all bookings at each lead time (up to 365 days).

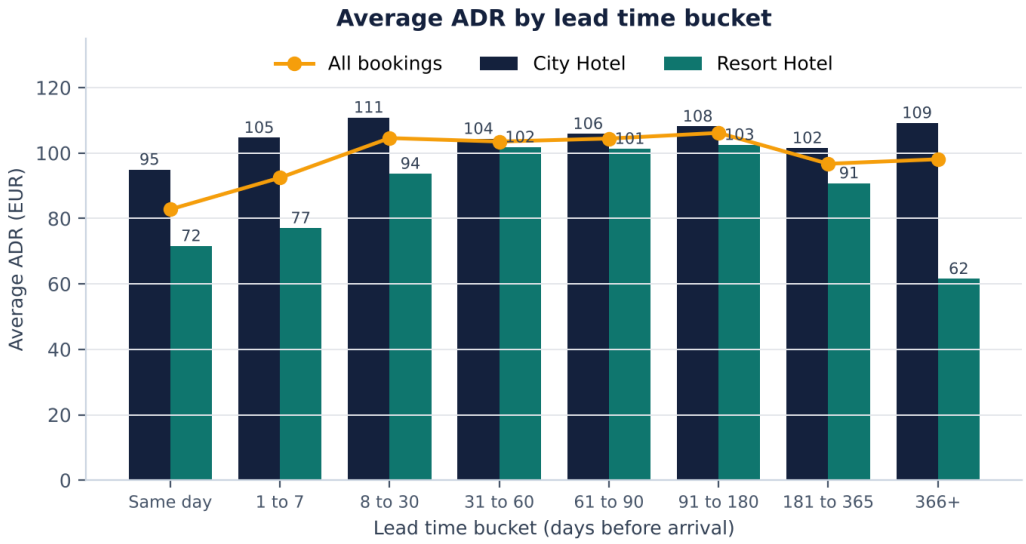


Figure 10. Average ADR by lead time bucket, by hotel and overall.

Lead time	Bookings	Share of bookings	Share of total ADR	Avg ADR
Same day	5,915	7.9%	6.5%	EUR 82.8
1 to 7 days	11,929	15.9%	14.7%	EUR 92.4
8 to 30 days	13,677	18.2%	19.0%	EUR 104.6
31 to 90 days	18,412	24.5%	25.5%	EUR 104.0
91 to 180 days	14,618	19.4%	20.6%	EUR 106.2
181+ days	10,615	14.1%	13.7%	EUR 96.8

Takeaway. The revenue pace curve has the same shape as the booking pace curve because ADR is similar across most lead times. Same-day bookings are the exception: EUR 82.8 on average against EUR 104.6 to 106.2 for bookings made 8 to 180 days ahead.

Source: Hotel Booking Demand dataset (Antonio, Almeida and Nunes, 2019)

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SECTION 10

Top 10 arrival dates

The ten busiest arrival dates each hold between 181 and 255 stayed bookings, and a single large block explains the peak on most of them.

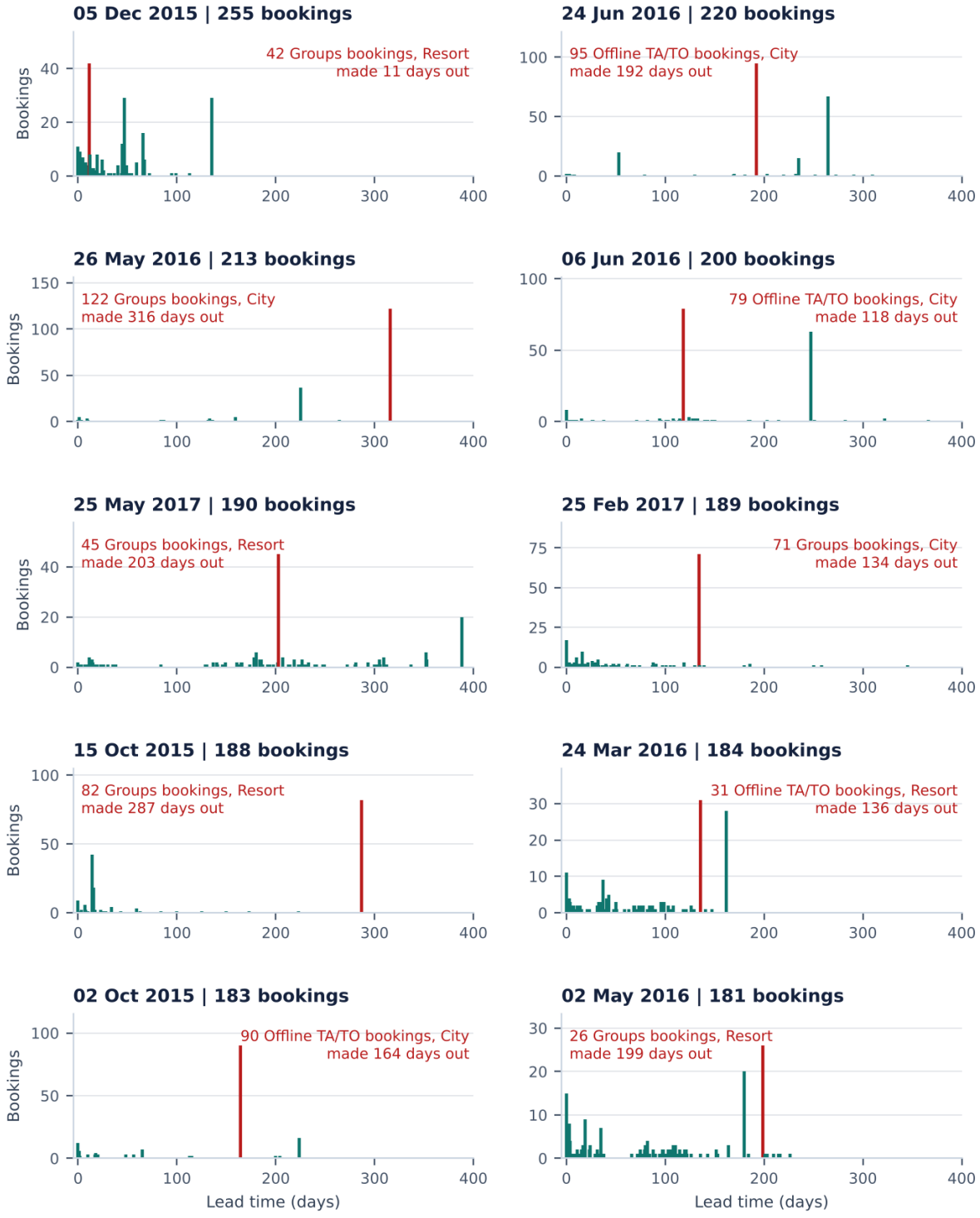


Figure 11. Bookings by lead time for each of the ten busiest arrival dates. The largest single-day block is highlighted in red. Axis shown to 400 days (one booking lies beyond).

Source: Hotel Booking Demand dataset (Antonio, Almeida and Nunes, 2019)

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SECTION 11

Top 10 arrival dates: block detail

On six of the ten dates one block made on a single day accounts for 38% to 57% of the date's bookings.

Arrival date	Bookings	Largest block	Segment	Hotel	Made	Share of date
5 Dec 2015	255	42	Groups	Resort / City	11 days out	16%
24 Jun 2016	220	95	Offline TA/TO	City	192 days out	43%
26 May 2016	213	122	Groups	City	316 days out	57%
6 Jun 2016	200	79	Offline TA/TO	City	118 days out	40%
25 May 2017	190	45	Groups	Resort	203 days out	24%
25 Feb 2017	189	71	Groups	City	134 days out	38%
15 Oct 2015	188	82	Groups	Resort	287 days out	44%
24 Mar 2016	184	31	Offline TA/TO	Resort	136 days out	17%
2 Oct 2015	183	90	Offline TA/TO	City	164 days out	49%
2 May 2016	181	26	Groups	Resort	199 days out	14%

- Every one of the ten largest blocks is a Groups or Offline TA/TO booking: six Groups and four Offline TA/TO.
- Blocks range from 26 to 122 bookings and were made between 11 and 316 days before arrival, most of them more than four months ahead.
- Together the ten blocks account for 683 of the 2,203 bookings on these dates (31%).
- The busiest dates are therefore not produced by an unusually strong stream of transient bookings, but by large contracted blocks placed on a single day.

Takeaway. Peak-date demand is largely a function of when group and travel-trade blocks are placed. Tracking those blocks separately from transient pickup explains most of the variation between the busiest dates.

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SECTION 12

Recommendations and conclusion

What the pace and pickup patterns mean for forecasting, pricing and inventory control.

1

Forecast by hotel and by segment

Median lead time runs from 3 days (Complementary, Aviation) to 107 days (Groups), and Resort books later than City. Separate pace curves capture these differences; a single blended curve does not.

2

Track block bookings separately

The ten busiest dates were each shaped by one Groups or Offline TA/TO block of 26 to 122 bookings. Monitoring block pickup on its own isolates the demand that transient forecasts miss.

3

Monitor the final week daily

23.7% of bookings arrive within 7 days and 7.9% on the day itself, mostly from Online TA, Direct and Corporate. This is where daily pickup tracking changes outcomes.

4

Plan weekend and weekday arrivals differently

Weekend arrivals book about ten days earlier (median 51 against 41). Weekday arrivals depend more on late pickup, with 9.4% booked on the day.

5

Protect rate on late demand

Same-day bookings average EUR 82.8 against EUR 104.6 to 106.2 for bookings made 8 to 180 days ahead. Late inventory should be priced with that gap in mind.

6

Use early pace for long-range planning

14.1% of bookings arrive more than 180 days ahead, mostly from Offline TA/TO, Online TA and Groups. Early pace is a usable signal for the following season.

Conclusion

Booking behaviour in this dataset is late-leaning but far from uniform. Half of stayed bookings are made within 45 days of arrival and nearly a quarter within a week, yet one booking in seven is made more than six months ahead. Resort demand arrives later than City demand, weekend arrivals book earlier than weekday arrivals, and the market segments span a window from 3 to 107 days. Rates are lowest on the arrival day. Most importantly, the busiest arrival dates are set by single large group and travel-trade blocks placed months in advance. Forecasting and revenue management that reflect these differences, by hotel, weekday, segment and block, will describe demand build far more accurately than a single aggregate pace curve.

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